

Stijn Denissen

FWO JUNIOR POSTDOC · VUB

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Professional Experience

- 2024-Now **Junior postdoctoral fellow FWO**, AIMS lab, VUB
Postdoctoral researcher funded by a personal junior postdoctoral fellowship. Title: Leveraging recent advances in explainable AI to decode cognitive functioning from T1/FLAIR weighted MR images in MS - An international federated learning approach (12A6U25N). More info: See 'Awards' tab.
- 2019-2024 **PhD Student (VLAIO Baekeland)**, Industrial doctorate. AIMS lab (VUB) and icometrix NV (Leuven)
Title: Structural brain damage and cognition in MS: an AI approach (HBC.2019.2579). See 'Education' and 'Awards' for more detail.
- 2018-2019 **Research assistant Prof. Guy Nagels**, AIMS lab, VUB
Initial lab experience working on various projects, including a project on NMSC Melsbroek rehabilitation data, resulting in a paper in the Journal of Central Nervous System Disease, and continuing the Cochrane update.
- 2018 **Physical therapist**, National Multiple Sclerosis Center Melsbroek
Rehabilitation of patients with multiple sclerosis.
- 2018 **Research assistant Prof. Geert Verheyden**, Dept. Rehabilitation Sciences, KU Leuven
Updating the 2013 Cochrane review of Prof. Verheyden, resulting in a paper in 2019.
- 2017 **Physical therapist**, GIRA Campus Pellenberg, UZ Leuven
Geriatric intensive rehabilitation, mainly of patients having suffered a stroke.

Education

Vrije Universiteit Brussel

Brussels

PHD MEDICAL SCIENCES

2019 - 2024

- Title: Structural brain damage and cognition in MS: an AI approach
- Promoters: Prof. Dr. ir. Guy Nagels, Dr. Diana Sima, Prof. Dr. ir. Jeroen Van Schependom
- Personal grant: VLAIO Baekeland (HBC.2019.2579)
- Defense date: 30/01/2024

KU Leuven

Leuven

MSC REHABILITATION SCIENCES

2015 - 2018

- Thesis: Trunk rehabilitation in the different recovery phases post-stroke: a systematic review and meta-analysis
- Promoter: Prof. Dr. Geert Verheyden
- Track: Neurorehabilitation
- Note: I was selected for a research internship on brain stimulation under supervision of Prof. Koen Cuypers.
- Grade: Cum Laude

KU Leuven

Leuven

BSC REHABILITATION SCIENCES

2012 - 2015

- Grade: Cum Fructu

Gymnasium Beekvliet

Sint-Michielsgestel

GYMNASIUM (HIGH SCHOOL)

2006 - 2012

Awards and grants

Over the years, I was successful in several research grants that mainly relied on the two research lines on brain age and federated learning from my PhD. The same research lines also resulted in 2 Chinese Scholarship Council (CSC) for PhD students Tianzheng Hu and Xinguang Wang (cfr. "mentoring").

2024	FWO Senior project , Fonds Wetenschappelijk Onderzoek (Role: Project design, writing + writing coordination)	€ 564.000
2024	PhD Thesis of the Year , VUB UMCOR	€ 600
2024	FWO junior postdoc , Fonds Wetenschappelijk onderzoek	± € 300.000
2023	Travel grant , ECTRIMS conference	€ 400
2022	FWO Travel grant , FWO (Role: Awardee)	€ 5.808
	IOF Proof of Concept , (Role: Project design, writing + writing coordination)	75.000
2021	Grant for OpenMR Virtual 2021 , Merck (Role: Awardee)	€ 1000
2019	Baekeland Grant , Vlaams agentschap innoveren & ondernemen (VLAIO)	€ 279.475
2019	Nominated: Best Young Scientist , Congress on Neurorehabilitation and Neural Repair	
2009	Enrichment project for gifted students , Gymnasium Beekvliet	

Teaching

I teach about clinical decision support systems (CDSS) to diverse clinical profiles including doctors, pharmacists, nurses and physiotherapists. My courses have a heavy emphasis on developing a strong background on artificial intelligence.

2026	Bachelor Thesis Medicine , Jury member: Jury member for Bachelor projects in the medicine track at VUB	Vrije Universiteit Brussel
2026	AI applications in medicine , Guest lecturer: Guest lecture entitled 'The Brain Clock and its neurological implications'. Please find the slideshow in the link below.	Vietnam National University
2025	Computational Neuroscience , Jury member: Jury member for Bachelor projects on machine learning on EEG data	Vietnam National University
2025	BICAMS training , Instructor: Teaching Bach Mai hospital students to take the BICAMS cognitive testing	Bach Mai hospital, Hanoi, VN
2020–Now	Medical Information- and Communication Systems , Assistant lecturer: Since 2020, I cover the aspect 'Clinical Decision Support Systems' with a heavy emphasis on providing a strong background in AI. Study Program: Master in Healthcare Management Course ID: 4012391ENW	VUB
2020–Now	Information systems , Assistant lecturer: Since 2020, I cover the aspect 'Clinical Decision Support Systems' with a heavy emphasis on providing a strong background in AI. Study Program: Master in Hospital Hygiene Course ID: B-KUL-E00I9A	KU Leuven
2019–2022	New therapeutic approaches to disorders of the CNS , Guest lecturer: Yearly guest lecture on the progress of my PhD research	VUB

Teaching and Supervision

Besides daily supervision, I coordinate our federated learning research line with a regular meeting. We align students and other researchers in the network to stay informed about each other's work and grant opportunities. I've also been a jury member of 2 master theses in the master in health care management (VUB).

PHD STUDENTS (CO-PROMOTER)

2025–Ongoing	Tianzheng Hu ({{P}}), Exploring the relationship between brain age and neurological disorders	Medical Sciences
2024–Ongoing	Xinguang Wang ({{P}}), Enhancing performance in deep learning-based brain age prediction models via federated learning	Medical Sciences

MASTER STUDENTS

Ongoing	Mohammad Waleed Adnan , explaining brain age models with XAI	Applied Computer Science
2024	Louise Van de Looverbosch , Age prediction based on vaginal microbiome profiles	Bio-engineering Science
2024	Fahimeh Heydarzadeh , Optimisation of a federated learning architecture for brainage	Applied Computer Science
2024	Alvaro Javier Vargas Guerrero , Federated learning in neuroimaging: pioneering predictive models for brain age estimation	Applied Computer Science
2023	André Vital Serafim Silva , Deep Learning for Brain Age Prediction from T1-Weighted Magnetic Resonance Images	Applied Computer Science
2022	Robin Lisa Van den Bosch , Does deep learning on imaging data enhance accuracy of brain age predictions?	Applied Computer Science
2022	Viktor-Jan De Deken , Evolutionary random subspace forest for brain age prediction in multiple sclerosis	Biomedical Science
2021	Siebe Clarebout , Modelling multimodal data to decode cognitive status in MS: A machine learning approach	Medicine

Experience abroad

2023	General Faculty Hospital, Prague (CZ) , Research Stay 3 months	Prague (CZ)
2017	Dr. Becker Kiliani-Klinik, Bad Windsheim (DE) , Internship 2 months	Bad Windsheim (DE)

Conferences

I presented at 7 international conferences in multiple sclerosis and neurorehabilitation, including two invited talks at the largest MS conference worldwide (ECTRIMS).

2025	ECTRIMS , Will AI shift decision making in favor of the PwMS in the next decade?	Barcelona (ES)
2025	BVNV , Artificiële intelligentie in MS	Blankenberge (BE)
2025	NNR , Surfing the smartphone wave to support clinical practice	Maastricht (NL)
2024	MOSA , Panel member	Maastricht (NL)
2024	ECTRIMS , Invited: What artificial intelligence tools are ready for MS clinical practice	Copenhagen (DK)
2022	IMSCOGS , Brain age as a surrogate marker for IPS in MS	Bordeaux (FR)
2021	ACTRIMS , Brain age in MS: An explainable principal component of brain MRI and potential sensitive cognitive biomarker	Virtual
2019	NNR , Interventions for preventing falls in people after stroke	Maastricht (NL)
2019	RIMS , The impact of cognitive dysfunction on locomotor rehabilitation potential in MS	Ljubljana (SL)
2019	RIMS , The effects of multidisciplinary rehabilitation in MS	Ljubljana (SL)

Extracurricular

RESEARCH COMMUNITY ENGAGEMENT

2025–now	iAIMS , Founder. iAIMS community. https://iaims.org	<i>International</i>
2025–now	AIMS lab , Supervisor. BIDS data format taskforce	<i>AIMS lab, VUB</i>
2021–2021	OpenMR , Organiser of the 2021 edition.. OpenMR conference. https://openmrbenelux.github.io/	<i>Virtual</i>

STUDENT ENGAGEMENT

2019–2021	NSE PhD Network , Board member. NSE PhD Network	<i>VUB</i>
2015–2017	Artifex , Co-founder. Artifex	<i>Leuven</i>
2014–2015	Apollooon , Board member. Apollooon	<i>KU Leuven</i>

Science communication and media

Over the years, I experimented with different forms of science communication, tailored for different audiences including physiotherapists, patients with MS, and even children. My idea to turn a paper into a song was particularly well received; SciMingo asked me to perform live at the final of the Flemish Thesis Prize at the Brussels city hall. Finally, I poured my PhD into a 3 minute song, performing it live when I received the ‘PhD of The Year’ award.

2026	AI x MS - The patient perspective , Brainstorm event at the National MS Center of Melsbroek, Belgium. People with MS met other clinicians, healthcare workers, researchers and the like to discuss 4 topics on AI in MS: 1. Which added value of AI do you perceive in MS care? 2. Which concerns do you have about AI in MS care? 3. How can we better involve people with MS in AI research? 4. What is the role of LLMs in the relationship patient - caregiver? ☒ More info in the LinkedIn post (Dutch)!	Event
2025	3 minute thesis competition Vietnam , Jury member for a national competition on science communication among engineering students. The challenge: to explain their research in 3 minutes with a single slide.	Event
2025	Vergeet Dementie , I contributed to the vulgarising book 'Vergeet Dementie' by authors Ms. Ceuleers and Prof. Engelborghs. I was interviewed by Ms. Ceuleers about the concept of 'brain age', and provided a layman summary for their book.	Event
2025	De Specialist , Interview on smartphone symptom screening together with Dr. Delphine Van Laethem (also in other media channels)	Interview
2024	LACTRIMS interview , Interview by Dr. Lorna Galleguillos on my talk entitled 'What artificial intelligence tools are ready for MS clinical practice' during the educational session on 'Clinical implications of digital monitoring and artificial intelligence in MS and related disorders' at ECTRIMS 2024 in Copenhagen	Interview
2024	Award ceremony 'PhD of The Year' , Live performance of my PhD in a 3-minute song, in synchrony with my 'a paper in a song' initiative.	Live Performance
2024	Het Nieuwsblad , Interview on brain age	Interview
2023	VUB PhD day , Live performance 'A paper in a song'	Live Performance
2023	Book presentation 'A question of truth' , Interview in the Brussels city hall together with my promoter Prof. Guy Nagels. We discussed our chapter contribution on medical diagnosis and AI.	Interview
2022	Award ceremony Flemish Thesis Prize (SciMingo) , Live performance 'A paper in a song'	Live Performance
2022	KNGF Fysiopraxis , Vulgarising article about our Cochrane review on interventions to prevent falls post-stroke. Title: 'Oefentherapie voorkomt vallen na een CVA'.	Article
2021	Children's university VUB , Title of the workshop I presented to primary school children: 'How old is my brain?'. In this workshop, I converted my PhD work in a comprehensible framework for young children. Together, we explored how AI can measure brain structures, and how we can use it to predict the age of my brain.	Event
2021	YouTube channel ,	Website
2021	Artsenkrant , Interview about ACTRIMS 2021 presentation on brain age.	Interview
2021	Wetenschap Brussel + EOS (Cognition) , Blog about cognition, together with Dr. Delphine Van Laethem. Title: 'Hersenspinsels: een editie over cognitie'	Article
2021	Haelio Neurology , Interview about ACTRIMS 2021 presentation on brain age	Interview
2020	Wetenschap Brussel + EOS (AI) , Blog about artificial intelligence. Title: 'Hersenspinsels: artificiële intelligentie'	Article

Note: Please click text in bold for more info if available. [Languages](#)

☒☒ Dutch	Native	☒☒ English	Fluent
☒☒ French	Proficient	☒☒ Italian	Proficient
☒☒ German	Proficient	☒☒ Czech	Basic

Publications

Van Laethem D, Costers L, Miravalle A, Alvarez E, Myers B, Boster A, Halawani HA, Dewil M, Cambron M, Grothe M, **Denissen S**,

- Bartz R, Sima D, Ribbens A, Smeets D and Nagels G 2026. The relationship between objective and subjective cognitive performance and clinical and MRI disease burden in early multiple sclerosis. *Scientific Reports* TBD. **IF: 3.9, Q1 2024**
- Denissen S**, Laton J, Grothe M, Vaneckova M, Uher T, Kudrna M, Horáková D, Baijot J, Penner I-K, Kirsch M, Motýl J, De Vos M, Chén OY, Van Schependom J and Nagels G 2026. Real-world federated learning for brain imaging scientists. *Frontiers in Digital Health* 8:1691088. **IF: 3.8, Q1 2024**
- Xinguang Wang, Nguyen Linh Trung, Oliver Y. Chen, Jeroen Van Schependom, Stijn Denissen and Guy Nagels 2026. PDD-FANs: Personalized Decentralized Federated Adversarial Networks with Defense Mechanism in Non-IID Setting. *IEEE TrustCom 2025 (Conference)*.
- Xinguang Wang, Nguyen Linh Trung, Oliver Y. Chen, Alvaro Javier Vargas Guerrero, Quang Manh Doan, Tianzheng Hu, Stijn Denissen, Jeroen Van Schependom and Guy Nagels 2026. Federated Adversarial Network (FAN): A Generative Proxy-Based, Defense-Enhanced, and Non-IID-Robust Federated Learning Method. *SSRN (preprint)*.
- Sosa A, O'Neill KA, Billiet T, Chen J, Couvreur L, **Denissen S**, Lustberg M, Pehel S, Ribbens A and Krupp L 2025. Incorporating measures of cognitive processing speed and brain volume in clinical management of pediatric onset MS. *Pediatric Neurology*. **IF: 2.1, Q2 2024**
- Trinh Ngoc Huynh, Nguyen Duc Kien, Nguyen Hai Anh, Dinh Tran Hiep, Manuela Vaneckova, Tomas Uher, Jeroen Van Schependom, Stijn Denissen, Tran Quoc Long, Nguyen Linh Trung and Guy Nagels 2025. Latent Representation Learning from 3D Brain MRI for Interpretable Prediction in Multiple Sclerosis. *arXiv preprint arXiv:2510.00051*.
- Denissen S***, Van Laethem D*, Baijot J, Costers L, Descamps A, Van Remoortel A, Van Merhaegen-Wieleman A, D'hooghe M, D'Haeseleer M, Smeets D, Sima DM, Van Schependom J and Nagels G 2025. A New Smartphone-Based Cognitive Screening Battery for Multiple Sclerosis (icognition): Validation Study. *Journal of Medical Internet Research* 27, e53503. **IF: 6.0, Q1 2024**
- De Keersmaecker E, Guida S, **Denissen S**, Dewolf L, Nagels G, Jansen B, Beckwée D and Swinnen E 2025. Virtual reality for multiple sclerosis rehabilitation. *Cochrane Database of Systematic Reviews* (1). **IF: 9.4, Q1 2024**
- Wittens MMJ*, **Denissen S***, Sima DM, ..., Engelborghs S 2024. Brain age as a biomarker for pathological versus healthy ageing—a REMEMBER study. *Alzheimer's research & therapy* 16.1 (2024): 128. **IF: 7.6, Q1 2024**
- Van Laethem D*, **Denissen S***, Costers L, Descamps A, Baijot J, Van Remoortel A, Van Merhaegen-Wieleman A, D'hooghe M, D'Haeseleer M, Smeets D, Sima DM, Van Schependom J and Nagels G 2024. The Finger Dexterity Test: Validation study of a smartphone-based manual dexterity assessment. *Multiple Sclerosis Journal* 30(1), 121-130. **IF: 5.0, Q1 2024**
- Denissen S**, Van Schependom J, Nagels G 2023. Medical diagnosis and a new kid in town: AI. Book chapter in VUB book 'A question of truth'.
- Thijs L, Voets E, **Denissen S**, Mehrholz J, Elsner B, Lemmens R and Verheyden G 2023. Trunk training for improving activities in people with stroke. *Stroke* 54(9), e427-e428. **IF: 7.9, Q1 2023**
- Thijs L, Voets E, **Denissen S**, Mehrholz J, Elsner B, Lemmens R and Verheyden G 2023. Trunk training for improving activities in people with stroke. *Cochrane Database of Systematic Reviews* (3). **IF: 8.8, Q1 2023**
- Baijot J, Van Laethem D, **Denissen S**, Costers L, Cambron M, D'Haeseleer M, D'hooghe MB, Vanbinst A, De Mey J, Nagels G and Van Schependom J 2022. Radial diffusivity reflects general decline rather than specific cognitive deterioration in multiple sclerosis. *Scientific Reports* 12(1), 21771. **IF: 4.6, Q2 2022**
- Denissen S** and Nagels G 2022. Artificial intelligence will change MS care within the next 10 years: Yes. *Multiple Sclerosis Journal* 28(14): e53503. **IF: 5.8, Q1 2022**
- Denissen S**, Engemann DA, De Cock A, Costers L, Baijot J, Laton J, Penner I-K, Grothe M, Kirsch M, D'hooghe MB, D'Haeseleer M, Dive D, De Mey J, Van Schependom J, Sima DM and Nagels G 2022. Brain age as a surrogate marker for cognitive performance in multiple sclerosis. *European journal of neurology* 29(10): 3039-3049. **IF: 5.1, Q1 2022**
- Denissen S**, Chen OY, De Mey J, De Vos M, Van Schependom J, Sima DM and Nagels G 2021. Towards multimodal machine learning prediction of individual cognitive evolution in multiple sclerosis. *Journal of personalized medicine* 11(12), 1349. **IF: 3.508, Q2 2021**
- Baijot J*, **Denissen S***, Costers L, Gielen J, Cambron M, D'Haeseleer M, D'hooghe MB, Vanbinst A, De Mey J, Nagels G and Van Schependom J 2021. Signal quality as Achilles' heel of graph theory in functional magnetic resonance imaging in multiple sclerosis. *Scientific Reports* 11(1), 7376. **IF: 4.997, Q2 2021**
- Denissen S**, Staring W, Kunkel D, Pickering RM, Lennon S, Geurts ACH, Weerdesteyn V and Verheyden GS 2020. Interventions for preventing falls in people after stroke. *Stroke* 51 (3), e47-e48. **IF: 7.914, Q1 2020**

Denissen S, De Cock A, Meurrens T, Vleugels L, Van Remoortel A, Gebara B, D'Haeseleer M, D'Hooghe MB, Van Schependom J, and Nagels G 2019. The Impact of Cognitive Dysfunction on Locomotor Rehabilitation Potential in Multiple Sclerosis. Journal of central nervous system disease 11, 1179573519884041.

Denissen S, Staring W, Kunkel D, Pickering RM, Lennon S, Geurts ACH, Weerdesteyn V and Verheyden GS 2019. Interventions for preventing falls in people after stroke. Cochrane database of systematic reviews (10). **IF: 7.89, Q1 2019**

* indicates joint first authorship.